VECTOR

3rdParty MCAL Integration

Technical Reference

NXP S32K3XX

Version 1.13.1

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Document Information

History

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jmaier	2023-02-02	1.13.1	Adding restrictions for Mcu and Port (2.2.1.5.2) Update Linker error (2.2.1.11)

Reference Documents

No.	Source	Title	Version
[1]	Vector	UserManual_3rdParty-MCAL-Integration.pdf	See Delivery
[2]	Vector	ScreenCast_McalIntegration_Tresos.pdf	See Delivery

Scope of the Document

This document contains information about the integration of 3rd Party MCAL into Vector software stack.

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MCAL Integration

1.1 Type of Integration

Basic Integration
Both configuration tools, EB tresos™ as well as Vector DaVinci Configurator, are used for configuration.

Recommended workflow:
Start initial configuration with EB tresos™, export it in AUTOSAR format and import it into Vector DaVinci Configurator. Generation and minor changes in configuration are done in Vector DaVinci Configurator.
For usage with Vector DaVinci Configurator 5 please refer to chapter 'Mixed configuration tool usage' in the document `UserManual_3rdParty-MCAL-Integration.pdf` [1].



Multimedia Link
For additional information please also refer to the Vector screen cast referenced in ScreenCast_McalIntegration_Tresos.pdf [2].

1.2 MCAL Location within SIP

The 3rd Party MCAL is separated from the Vector parts within the SIP. It might not even be part of the delivery. Please refer to chapter 'First Steps' in document `UserManual_3rdParty-MCAL-Integration.pdf` [1].

1.3 Supported 3rd Party Products

This integration supports the following NXP targets:

1.3.1 S32K2xx

1.3.1.1 ASR4.3

- S32K2XX_MCAL4_3_EAR_0_8_1 (Base Package),
  S32K2XX_Crypto_MCAL4_3_EAR_0_8_1 (Security Package)

1.3.2 S32K3xx / MWCT2x

1.3.2.1 RTD

1.3.2.1.1 ASR4.7/R21-11

- SW32K3XX_RTD_R21-11_3.0.0_CD04
- SW32K358_RTD_R21_11_3.0.0_CD03_P01_D2211
- SW32K358_RTD_R21-11_3.0.0_CD03

1.3.2.1.2 ASR4.4

- SW32K396_RTD_4.4_3.0.0_CD02 incl. P01
- SW32K3_RTD_4.4_2.0.2
- SW32K3_RTD_4.4_2.0.1
- SW32K3_RTD_4.4_2.0.0

1.3.3 S32K3xx

1.3.3.1 RTD

1.3.3.1.1 ASR4.4

- SW32K3_RTD_4.4_2.0.0_CD01
- SW32K3_RTD_4.4_1.0.0
- SW32K3_RTD_4.4_1.0.0_CD01
- SW32K3_RTD_4.4_0.9.0
- SW32K3_RTD_4.4_0.8.1 incl. FBR 1.1
- SW32K3_RTD_4.4_0.8.0_P02 (incl. Crypto)
- SW32K3_RTD_4.4_0.8.0

1.3.3.2 SPD

1.3.3.2.1 ASR4.4

- S32K396_SPD_0.8.0
- S32K3_SPD_1.0.1
- S32K3_SPD_1.0.0

1.3.3.3 SBC

1.3.3.3.1 ASR4.4

- FS26_4.4_1.0.0
- FS26_B0_4.4_1.0.0_CD01
- FS26_4.4_0.9.1



Note
Please refer to the Release Notes of the 3rd Party Products for further information, e.g. regarding supported versions, derivatives and compilers.



Note

Only official 3rd Party vendor releases are part of this Vector integration package. Therefore, any customer-specific releases cannot be considered.



Caution
To find out if there are further Hotfixes available for your MCAL package, please contact the 3rdParty Vendor.

It is essential to replace the affected EB tresos™ module plugins in your original package before you start Script_MCAL_Prepare.bat.

1.4 Configuration Tools

- Vector DaVinci Configurator 5
- EB tresos™

Vector Comment

The user should consider the attached `UserManual_3rdParty-MCAL-Integration.pdf` [1] for further information regarding Vector integration and setup of a project.

2.1 Known Issues

The MCAL package in use might not be the latest one. Updates or Hot Fixes might be available from the vendor.

If you discover errors in the MCAL during development or suspect that faulty behavior could be caused by the MCAL, please contact the 3rdParty Vendor. If there are updates for your MCAL available, it might be helpful to use them. The corresponding 3rdParty Vendor Release Notes will tell you which errors have been fixed.

The errors documented in the following chapters were detected at Vector during the MCAL integration and reported to the 3rdParty Vendor.



Note
Vector makes every effort to ensure that this integration is compatible with all MCAL packages of the corresponding Controller Family as long as the 3rdParty Vendor does not change the structure of the packages. However, not all MCAL packages can be tested and therefore the user may find errors either in Vector's files or in the MCAL that are not listed in this document.



Caution
Modifications to the MCAL, which may be described in the following chapters under the keyword 'Workaround', must take place after the 3rdPartyMCALIntegrationHelperTool has run. Otherwise the changes will be overwritten.



Note
Necessary patches are provided via the SIP folder ThirdParty\<Short_Name>\VectorIntegration\Patches\ (called 'patch folder' in this chapter).

2.1.1 S32K2xx

2.1.1.1 S32K2XX_MCAL4_3_EAR_0_8_1 / S32K2XX_Crypto_MCAL4_3_EAR_0_8_1

2.1.1.1.1 Usage of the security package

If the Security package of the MCAL shall be used, the following changes have to be done:



Workaround
Merge the Crypto plugin from the security package with the basic MCAL package. Replace all modules in the plugins folder of the MCAL base package with the corresponding modules of the security package.

2.1.1.1.2 Usage of different Wdg instances

The pre-configured EcuC settings allow only to use one Wdg instance (SWT0). The following workaround can be used for other Wdg instances:



Workaround
In /MICROSAR/EcuC/EcucGeneral/BswInitialization/InitFunction, duplicate Wdg_43_Instance0_Init and adapt the settings depending on the chosen implementation variant.

2.1.1.1.3 Wdg_CfgExt.h not generated

The file Wdg_CfgExt.h is not generated leading error messages of the compiler. There is no workaround available.

This issue was reported to NXP with the ticket ID 00280371.

2.1.2 S32K3XX

2.1.2.1 SW32K3_RTD_4.4_0.8.0 / SW32K3_RTD_4.4_0.8.0_P02

2.1.2.1.1 ISR function not found

Compiling can lead to the following error message:

Swt_Ip_Irq.c, 107: ISR(Swt_Ip_Swt0_Isr); - explicit type is missing ("int" assumed)

- similar errors like this in Adc_Sar_Ip_Isr.c and Bctu_Ip_Isr.c

This issue was reported to NXP with ticket number ID 00311377.

2.1.2.1.2 Issues with memory sections in Mcl and Spi module

There are two issue regarding memory sections in the Spi module:

1. In Spi_IPW.c, the memory section “SPI_START_SEC_CODE”, but not closed.

2. In Lpspi_Ip.h, there is a memory section defined with
   #define SPI_START_SEC_CODE
   #define SPI_STOP_SEC_CODE
   But the includes of the Spi_MemMap.h are missing.

3. In Lpspi_Ip.c, there is a #define SPI_STOP_SEC_CODE without the #include "Spi_MemMap.h" and the SPI_START_SEC_CODE is missing.
4. In Emios_Common.h, there are #define MCL_START_SEC_CODE and #define MCL_STOP_SEC_CODE without MemMap includes.

Both issues were reported to NXP with ticket number ID 00311824.

2.1.2.1.3 Compiling issue of Spi module

Compiling the Spi module can lead to the following error message:

1) “Lpspi_Ip_Types.h: identifier "bool" is undefined” and
2) “Spi.c, 2127: identifier "u32SpiCoreID" is undefined”.

1) Boolean should be used instead of bool.
2) #if ((SPI_SLAVE_SUPPORT == STD_ON) && (SPI_LEVEL_DELIVERED != LEVEL0)) should be
#if (((SPI_SLAVE_SUPPORT == STD_ON) && (SPI_LEVEL_DELIVERED != LEVEL0)) \
|| (SPI_DEV_ERROR_DETECT == STD_ON))
This issue was reported to NXP with ticket number ID 00311377.

2.1.2.1.4 Compiling error in OsIf_Timer.c

Compiling the file “OsIf_Timer.c” leads to the error message

"identifier uin32_t is undefined"

Because uin32_t should be uint32 instead.
This issue was reported to NXP with ticket number ID 00311836.

2.1.2.1.5 Missing compiler switch for intermodule check in Qspi_Ip.h

The compiler switch for the intermodule check between Qspi and StandardTypes are missing leading to a compiler error.
This issue was reported to NXP with ticket number ID 00311040.

2.1.2.1.6 Configuration problems with Icu module – missing reference to DMA

IcuDMAChannelRef needs a reference to a DMAChannel, but the MCL module does not have a MclConfigSet container or a DMAChannel.

This issue was reported to NXP with ticket number ID 00309782.



Workaround
In DaVinci Configurator 5, set the reference to the DMAChannel manually and set it as user defined.

2.1.2.1.7 Compiling issue of Icu module

Compiling the Icu module can lead to the following error message:
const Icu_Ipw_IpConfigType Icu_Ipw_IpConfig_PB[0U] =

```
                                    ^
"D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41032\internal\StartApplication\Appl\GenData\src\Icu_Ipw_PBcfg.c",145  Error[Pe094]:
          the size of an array must be greater than zero
```

This issue was reported to NXP with ticket number ID 00311858.

2.1.2.1.8 Compiling issue of Mcl module

Compiling the Mcl module can lead to the following error message:
Emios_Common.c: Emios_Common_Config = NULL - identifier "NULL" is undefined.

This issue was reported to NXP with ticket number ID 00311377.

2.1.2.1.9 Compiling issue of Mcu module

Compiling the Mcu module can lead to the following error message: MC_ME.c: identifier "Power_Ip_MC_ME_Type", "MC_ME_MODE_UPD_MODE_UPD_MASK" and "MC_ME_PRTN0_PUPD_PCUD_MASK" is undefined.

The following compiler option should be used:
-DS32K3XX=1
-DS32K344=1

2.1.2.1.10 Compiling issue of Ocu module

Compiling the Ocu module can lead to the following error message:

```
};
  ^
"D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41032\internal\StartApplication\Appl\GenData\src\Ocu_PBcfg.c",99  Error[Pe029]:
          expected an expression
```

This issue was reported to NXP with ticket number ID 00311867.

2.1.2.1.11 Configuration problems with Pwm module

While trying to generate the module with DaVinci Configurator, the following error appears: (1803) Invalid XPath-expression for Attribute "ENABLE" of node "PwmEmios": (35016) Cannot get a resource file for target CORTEXM and derivate S32K3XX.

This issue was reported to NXP with ticket number ID 00309782.



Workaround
Switch of the “Performance Optimization” in the Generator user settings and exclude the modules with references to Emios resp. Pwm (e.g. Pwm, Mcl, Base) from the generation process.

Use the generation result of Tresos for the other modules.

2.1.2.1.12 Compiling issue of Pwm module

Compiling the Pwm module can lead to the following error message:
Pwm_Cfg.h, 281: PWM_CONFIG_PB - identifier "Pwm_ChannelConfigType" is undefined

This issue was reported to NXP with ticket number ID 00311377.



Workaround
In DaVinci Configurator 5, switch the “Implementation Variant” of the Pwm module to Precompile and generate again.

2.1.2.1.13 Compiling issue of Wdg module

Compiling the Wdg module can lead to the following error message:

- Wdg_Channel.c, 962: if((uint32)STD_ON == Wdg_E_Mode_Failed.state) - identifier "Wdg_E_Mode_Failed" is undefined

- same in line 972, 1086 and 1094

This issue was reported to NXP with ticket number ID 00311377.



Workaround
In DaVinci Configurator 5, uncheck the option “Disable Dem Report Error Status” (in Wdg/WdgGeneral/AutosarExt) and generate again.

2.1.2.1.14 Missing compiler switch for intermodule check in hse_ip.h

The compiler switch for the intermodule check between Hse_Ip.h and StandardTypes are missing. This leads to an error message of the compiler.
This issue was reported to NXP with ticket number ID 00314389.

2.1.2.1.15 Invalid redeclaration of type name

Compiling leads to the following error messages:

"D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41653\external\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_0.8.0_P02\eclipse\plugins\Base_TS_T40D34M8I0R0\include\BasicTypes.h",52
Error[Pe256]: invalid redeclaration of type name "uint32_t" (declared at line 80 of "D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41653\external\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_0.8.0_P02\eclipse\plugins\Crypto_TS_T40D34M8I0R0\examples\EBT\Crypto_Example_001\include\std_typedefs.h")
"D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41653\external\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_0.8.0_P02\eclipse\plugins\Base_TS_T40D34M8I0R0\include\BasicTypes.h",100
Error[Pe256]: invalid redeclaration of type name "int32_t" (declared at line 79 of "D:\usr\usage\Delivery\CBD20x\CBD2000733\DELIVERY41653\external\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_0.8.0_P02\eclipse\plugins\Crypto_TS_T40D34M8I0R0\examples\EBT\Crypto_Example_001\include\std_type defs.h")

"uint32_t" and "int32_t" are defined in both BasicTypes.h of the Base module and the std_typedefs.h which is needed of HSE (see restrictions chapter 2.2.1.7)

This issue was reported to NXP with ticket number ID 00314388.

2.1.2.2 SW32K3_RTD_4.4_0.8.1 incl. FBR 1.1

2.1.2.2.1 Invalid redeclaration of type name

See chapter 2.1.2.1.15.

2.1.2.2.2 Configuration problems with Pwm module

See chapter 2.1.2.1.11.

2.1.2.2.3 Missing compiler switch for intermodule check in Wdg

The compiler switch for the intermodule check between Wdg_Channel.c and SchM_Wdg.h, Wdg_ChannelType.h and WdgIf.h, Wdg_Cfg_Defines.h and Wdg_If_types are missing. This leads to an error message of the compiler.
This issue was reported to NXP with ticket number ID 00321171.

2.1.2.2.4 Issue with ISR in Adc

The compiling of Adc leads to the error message:

```
"../../../external/ThirdParty/Mcal_S32x/Supply/SW32G2_RTD_4.4_0.9.0/eclipse/plugins/Adc_TS_T40D11M9I0R0/src/Adc_Sar_Ip_Isr.c", line 41: warning #837-D:
          omission of explicit type is nonstandard ("int" assumed)
  ISR(Adc_Sar_0_Isr); /* IAR requires prototypes. */
  ^

"../../../external/ThirdParty/Mcal_S32x/Supply/SW32G2_RTD_4.4_0.9.0/eclipse/plugins/Adc_TS_T40D11M9I0R0/src/Adc_Sar_Ip_Isr.c", line 41: error #65:
          expected a ";"
  ISR(Adc_Sar_0_Isr); /* IAR requires prototypes. */
```

The included file “OsIf_Internal.h” from the Autosar Adaptive module OsIf includes “Os.h” for the define of ISR if “USING_OS_AUTOSAROS” is defined. But OsIf_Cfg.h is not included so that “USING_OS_AUTOSAROS” is never defined.

This issue was reported to the NXP ticket system with the case number 00321171.



Workaround
According to NXP support, OsIf_Internal.h is only a stub file and the missing
#include “OsIf_Cfg.h”
Can be added manually.

2.1.2.2.5 SystemTimer not defined in OsIf_Cfg.c

Compiling OsIf_Cfg.c from the Autosar Adaptive module OsIf leads to the following error message:

Osif_Cfg.c, line 136: error #20 identifier "SystemTimer" is undefined.

SystemTimer is defined in Os, but Os is not included in "OsIf_cfg.c"

This issue was reported to the NXP ticket system with the case number 00321171.



Workaround
According to NXP support, OsIf_cfg.c is only a stub file and the missing
#include “Os.h”
Can be added manually.

2.1.2.2.6 Compiling error of Pwm module

Compiling the Pwm module leads to following error message:

"../../../external/ThirdParty/Mcal_S32k/Supply/SW32K3_RTD_4.4_0.8.1/eclipse/plugins/Pwm_TS_T40D34M8I1R0/src/Pwm.c", line 197: error #144:
a value of type "void *" cannot be used to initialize an entity of
type "boolean"
static Pwm_DriverStateType Pwm_aState[1U] = PWM_DRIVER_STATE_INITIALIZATION;

This issue was reported to the NXP ticket system with the case number 00321171.

2.1.2.2.7 Include of nvic.h

Emios_Pwm_Ip.c includes the file nvic.h which is not needed.
This issue was reported to the NXP ticket system with the case number 00321171.



Workaround
According to NXP support, the include can be deleted.

2.1.2.2.8 Issue with McuResetReasons

The McuResetReasons are generated as following:
PowerIpConf_PowerIpResetReasonConf_POWER_IP_POWER_ON_RESET

The Vector EcuM references it with the naming convention according to Autosar:
McuConf_McuResetReasonConf_MCU_POWER_ON_RESET

This leads to error messages like this:

```
if ((EcuM_GetReasonOfWakeupSourceList(loopCount)) == resetReason)
             ^
".\BSW\EcuM\EcuM.c",3305  Error[Pe020]:
        identifier "McuConf_McuResetReasonConf_MCU_POWER_ON_RESET" is
        undefined
```

This issue is discussed with NXP for a similar device in ticket #00326263.

2.1.2.3 SW32K3_RTD_4.4_1.0.0

2.1.2.3.1 Linker error using IAR compiler

Using IAR compiler can lead to linker errors like:

"PRAM0_BLOCK0__MSR_VAR_CLEARED": place in PRAM0_BLOCK0
^
"G:\internal\StartApplication\Appl\Source/vLinkGen_Template.icf",188 Error[Lp005]:
placement includes a mix of sections with content (example
"rw data section .MSR_VAR_CLEARED in Icu.o") [suitable for placement
in ROM] and sections without content (example
"zi section .MSR_VAR_CLEARED in Adc.o symbols: [Adc_pCfgPtr]")
[suitable for placement in RAM]

This issue was reported to the NXP ticket system with the case number 00449194.



Workaround
According to NXP support, the RTD was tested with the IAR compiler option
`--do_explicit_zero_opt_in_named_sections`
instead of
`--no_explicit_zero_opt`
as mentioned in the release notes.

2.1.2.4 SW32K3_RTD_4.4_2.0.0_CD01

2.1.2.4.1 Port: missing MemMap includes in Port.h

In Port.h, a memory section is defined, but the includes of Port_MemMap.h is missing. This leads to compiler errors.

This issue was reported to the NXP ticket system for a similar device with the case number 00441486.

2.1.2.4.2 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3 S32K3xx / MWCT2x

2.1.3.1 SW32K3_RTD_4.4_2.0.0

2.1.3.1.1 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3.1.2 Port: Memory section no closed in Port_Ipw.c

In the file Port_Ipw.c, a memory section is opened, but not closed at the end.

This issue was reported to the NXP ticket system for a similar device with the case number 00458192.

2.1.3.2 SW32K3_RTD_4.4_2.0.1

2.1.3.2.1 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3.2.2 Port: Memory section no closed in Port_Ipw.c

See chapter 2.1.3.1.2.

2.1.3.3 SW32K3_RTD_4.4_2.0.2

2.1.3.3.1 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3.3.2 Port: Memory section no closed in Port_Ipw.c

See chapter 2.1.3.1.2.

2.1.3.4 SW32K396_RTD_4.4_3.0.0_CD02 incl. P01

2.1.3.4.1 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3.4.2 Mcl: start of DaVinci Configurator 5 blocked

In the file Mcl_s32k396_289bga.epd, the paramater trgmuxLogicTrigger_Input has the configuration possibilities TRGMUX_IP_INPUT_LOGIC0_VSS and TRGMUX_IP_INPUT_LOGIC1_VDD twice. This blocks the start of our DaVinci Configurator.

This issue was reported to the NXP ticket system for a similar device with the case number 00480978.

2.1.3.5 SW32K3XX_RTD_R21-11_3.0.0_CD04

2.1.3.5.1 Linker error using IAR compiler

See chapter 2.1.2.3.1.

2.1.3.5.2 Crypto: developed according to ASR 4.4

According to the release notes, Crypto is developed according to ASR 4.4 instead of ASR 21-11. Hence, in CryIf and CSM must be configured according to ASR 4.4 in DaVinci.

2.1.3.5.3 Crypto: missing defines for S32K311

According to the Crypto User Manual, S32K311 is supported by the Crypto driver. But in `Hse_Ip_Cfg.h`, the corresponding include is not generated. This leads to missing defines during the build process, e.g.

```
identifier "MU_Type" is undefined

  static inline void Mu_Ip_SetTxRegister(MU_Type *base, uint8
u8Channel, uint32 u32Value)
```

This issue was reported to the NXP ticket system with the case number 00510624 and NXP confirmed that Crypto is not supported by the MCAL.

2.1.3.5.4 Eth: possible Type mismatch in Eth_DataType

The Vector stack defines `Eth_DataType` as `uint32`, but NXP's Eth driver needs it as an `uint8`. Both is compliant to ASR, but leads to incompatibilities.

This issue was reported to the NXP ticket system with the case number 00510537.

2.1.4 SBC

2.1.4.1 FS26_4.4_0.9.1

2.1.4.1.1 Wdg_43_fs26: Issue in Tresos GUI

Adding Wdg_43_fs26 to the Tresos project can lead to error messages like:

```
Invalid XPath-expression for Attribute "ENABLE" of node "<a
href="datamodel://CLT_S32K3_SBC_SPD?ASPath:/Wdg/WdgDemEventParamet
erRefs">ASPath:/Wdg/WdgDemEventParameterRefs</a>": (1818) No value
found for object ""
```

The generation on the commandline is possible, anyway.



Workaround
Configure and generate the plugin in DaVinci Configurator 5 only.

2.1.4.2 FS26_B0_4.4_1.0.0_CD01

2.1.4.2.1 Sbc: nested memory sections

The file `sbc_spi_external_access.c` has two nested memory sections.

This issue was reported to the NXP ticket system for a similar device with the case number 00481793.

2.2 Restrictions

2.2.1 General

2.2.1.1 Missing header-file StandardTypes.h

Compiling is not successful since the header-file StandardTypes.h is missing.



Workaround
Create a stub file “StandardTypes.h” with the following content:

```
#include "Std_Types.h"
```

2.2.1.2 Missing header-file EcuM_Externals.h

Compiling is not successful since the header-file EcuM_Externals.h is missing.



Workaround
Create a stub file “EcuM_Externals.h” with the following content:

```
#include "EcuM_Cbk.h"
```

2.2.1.3 Missing header-file PlatformTypes.h

Compiling is not successful since the header-file PlatformTypes.h is missing.



Workaround
Create a stub file “PlatformTypes.h” with the following content:

```
#include "Platform_Types.h"
```

2.2.1.4 Missing header-file ComStackTypes.h

Compiling is not successful since the header-file ComStackTypes.h is missing.



Workaround
Create a stub file “ComStackTypes.h” with the following content:

```
#include "ComStack_Types.h"
```

2.2.1.5 Initialization

2.2.1.5.1 ConfigPtr Values in Config Variants

According to Autosar, in Config Variants VariantPreCompile and VariantLinkTime with only one configuration variant, the configuration pointer for the Initialization functions shall have NULL_PTR values. This is the default configuration in DaVinci Configurator 5. If a different ConfigPtrName is needed, it must be User defined and configured manually.

If Config Variant VariantPostBuild or multiple configuration variants are used, the configuration pointer must be different from NULL_PTR (AUTOSAR_SWS_BSWGeneral.pdf).

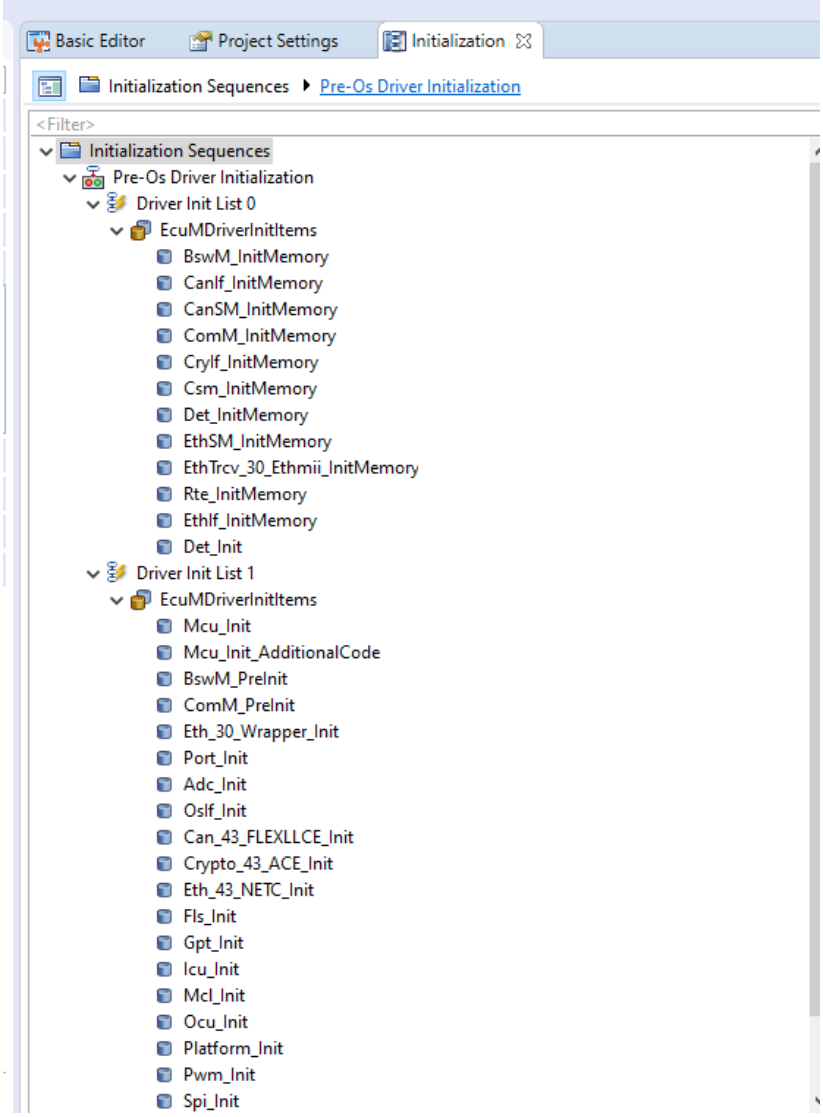
The naming convention according to Autosar is:

<Mip>_Config resp. <Mip>_Config_<PredefinedVariant.shortName> if different variants are used (AUTOSAR_TPS_ECUConfiguration.pdf).

The default value in DaVinci Configurator 5 is <Mip>_Config e.g. Mcu_Config or Spi_Config. If a different ConfigPtrName is needed, it must be User defined and configured manually.

2.2.1.5.2 Init order of Mcu and Port

To support the Eth start-up, the Port module is initialized before Mcu. If another order is needed by the user, this must be changed manually by the user in the “Initialization”-window:



Reference
Integration Manual for S32K3 ETH Driver
RTD_ETH_IM.pdf

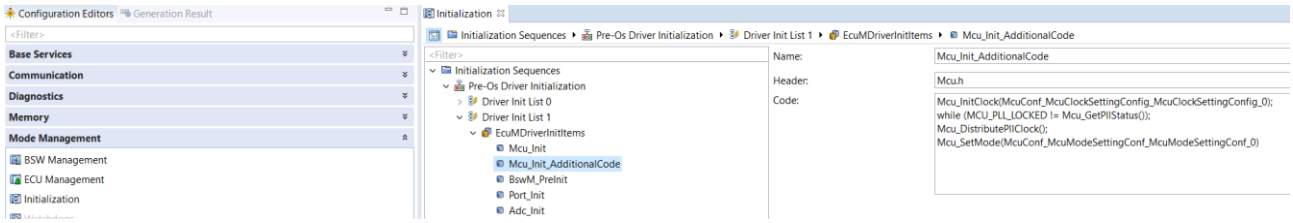
2.2.1.5.3 Mcu Init additional Code

The Mcu Init additional code cares for the initialization process as described in Nxp's Mcu Integration Manual. The function parameters

`McuConf_McuClockSettingConfig_McuClockSettingConfig_0`

`McuConf_McuModeSettingConf_McuModeSettingConf_0`

fit to the default value of the MCAL. If the configuration deviates from the default value or the optional initialization of Ram sections is required, this must be configured by the customer manually in the initialization settings:



2.2.1.6 Usage of Wdg Instances

The Wdg instances do not use VendorId and ApiInix for the exclusive areas / internal behavior. This leads to the restrictions in Rte / SchM if only Instance1 or Instance2 shall be used:

After consulting NXP with ticket number #00414546, a file "SchM_Wdg.h" shall be created which includes e.g. "SchM_Wdg_43_Instance1.h" and redefines the exclusive areas, e.g.

```
#define SchM_Enter_Wdg_WDG_EXCLUSIVE_AREA_00()
SchM_Enter_Wdg_43_Instance1_WDG_EXCLUSIVE_AREA_00()

#define SchM_Exit_Wdg_WDG_EXCLUSIVE_AREA_00()
SchM_Exit_Wdg_43_Instance1_WDG_EXCLUSIVE_AREA_00()
```

2.2.1.7 Crypto driver with (RTD) and without (MCAL) HSE interface

The usage of the Crypto driver of the Real Time Drivers (RTD) needs files from the Hardware Security Engine which can be downloaded from NXP. In this integration, the HSE firmware package `HSE_FW_S32K3XX_EAR_0_0_8_3` was used. The path is written to `Mcal_S32k_rules.mak` by the Mcal Integration Helper.

If the example files from the Crypto module are used, the path must be adopted to the use case in `Mcal_S32k_rules.mak`, e.g.

```
CC_INCLUDE_PATH += $(CRYPTO_BASE_PATH)\examples\EBT\
Crypto_Cmac_FastCmac\include
```

The usage of the HSE firmware package resp. example files can switched on and off in `Mcal_S32k_rules.mak`:

`EXCLUDE_HSE_PACKAGE = 0` → usage of HSE firmware package

`EXCLUDE_HSE_PACKAGE = 1` → usage of example files

If other MCAL packages, which do not need the HSE firmware package or example files, are used, the HSE sections shall be deleted in `Mcal_S32k_rules.mak`.

2.2.1.8 Crypto open source file "hse_srv_key_import_export.h"

Compiling can lead to the following error message:

"../../../external/ThirdParty/Mcal_S32k/Supply/SW32K3_RTD_4.4_0.8.1/eclipse/plugins/Crypto_TS_T40D34M8I1R0/examples/S32DS/Crypto_Example_001/include/hse_interface.h", line 49: fatal error #2330:
cannot open source file "hse_srv_key_import_export.h"
#include "hse_srv_key_import_export.h"

This is caused by long path names.



Workaround
Substitute the path as long as possible. In the example above, everything before external can be substituted.

2.2.1.9 Different generation results in Mcu module

Since EB tresos™ uses the index for the generation order in GUI mode, but alphabetical order on the command line, this can lead to generation differences of EB tresos™ and DaVinci, which calls EB tresos™ via commandline. This can occur e.g. for McuPeripherals if more than 10 resp. 100 are available.



Workaround
In EB tresos™, rename

McuPeripheral_0, McuPeripheral_1, … McuPeripheral_9

To

McuPeripheral_00, McuPeripheral_01, … McuPeripheral_09 if more than 10 are available, resp.

McuPeripheral_000, McuPeripheral_001, … McuPeripheral_099 if more than 100 are available.

Import the configuration to DaVinci Configurator Classic.

2.2.1.10 Additional compiler specific defines

For compiling, the MCAL needs additional compiler specific defines which are not part of the Vector files. As a workaround they are added to the example file _Compiler_Cfg.h. The user must take care that the right ones are used.

2.2.1.11 Linker error: Argument list too long

After it has compiled successfully, the linker shows the following error message in some environments:

```
LD          TestSuit.elf
sh: D:/uti/GHS/ARM/comp_202114/ccthumb: Argument list too long
make: *** [..\..\MakeSupport/Global.Makefile.target.4.mk:948: TestSuit.elf] Error 126
**********************************************************************************
```



Workaround

If you use the Vector make support, enable the usage of the response file in the makefile:

```
USE_RESPONSE_FILE_FOR_LINKAGE=1
```

2.2.2 MWCT2x

2.2.2.1 SW32K3_RTD_4.4_2.0.0

2.2.2.1.1 Support of MWCT2x

According to the release notes, the RTD supports also MWCT2x derivatives. How MWCT2x derivatives can be mapped to S32K3x derivatives, is described in the RTD release notes.

Glossary and Abbreviations

3.1 Glossary

Term	Description
3rd party components / MCAL	BSW modules not provided by Vector. Vector may have integrated the software within the SIP but does not take over any responsibility regarding functionality of these modules.
DaVinci Configurator	Configuration and generation tool for Vector MICROSAR components

Table 3-1 Glossary

3.2 Abbreviations

Abbreviation	Description
MCAL	Microcontroller Abstraction Layer
AUTOSAR	Automotive Open System Architecture
SIP	Software Integration Package (as provided by Vector)
Msn	Module Short Name according AUTOSAR

Table 3-2 Abbreviations

Contact

Visit our website for more information on

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